DA3409 Quiz 2 - Marking scheme (for TAs)

General: give marks for correct working even if a later number has a small slip. Do not take marks off twice for the same mistake: if a wrong answer from an earlier question is used correctly later, give full marks for the later step (error carried forward). Answers in Hindi, Tamil or any other language are marked the same way.

Q1 [1 mark]: (c). All or nothing.

Q2 [1 mark]: (b) 73,856 = 3 x 3 x 64 x 128 + 128. All or nothing.

Q3 [3 marks]
1 mark: uses out = (W - K + 2P)/S + 1 with the right values (W=64, K=5, P=2, S=2).
1 mark: output shape 32 x 32 x 16 (all three numbers needed).
1 mark: parameters = 5 x 5 x 3 x 16 + 16 = 1,216.

Q4 [3 marks]
1 mark: flattened size 32 x 32 x 16 = 16,384 (or the correct product of the student's own Q3 shape).
1 mark: weights = flattened size x 10.
1 mark: adds the 10 biases; total 163,850 (or the correct total from the student's own Q3 shape).

Q5 [4 marks]
1 mark: sets up the chain rule dL/dw = dL/dy . dy/dz . dz/dw with z = wx + b.
1 mark: dL/dy = (y - t).
1 mark: dy/dz = sigmoid(z)(1 - sigmoid(z)) = y(1 - y).
1 mark: final answer dL/dw = (y - t) y (1 - y) x.

Q6 [4 marks]
1 mark: during training each unit is set to zero at random with probability p.
1 mark: so a different thinned network is trained at each step and units cannot co-adapt (rely on particular other units).
1 mark: at test time no unit is dropped and activations are scaled by (1 - p) (or, with inverted dropout, scaled by 1/(1 - p) during training instead).
1 mark: why it reduces overfitting: it works like averaging an ensemble of many thinned networks (a regulariser). Just saying "it reduces overfitting" does not earn this mark.

Q7 [4 marks]
1 mark: exponentiates and divides by the sum of the exponentials.
1 mark: subtracts max(x) before exponentiating.
1 mark: the rest of the code is correct and returns the list of probabilities (a missing import is not penalised).
1 mark: explains that softmax(x - c) = softmax(x) because the factor e^(-c) cancels in the numerator and denominator.